Lesson Plan: How Do Robots Find Their Way?

Game	Doodlebot – The Lost Kingdom (autonomous navigation)
Audience	Grade 5 (ages 10 – 11); students already know what AI means
Duration	30 minutes

Learning Objectives: students will be able to…

1. **Explain** that a robot only knows what its sensors detect, and builds its own map in which unknown places are black.
2. **Describe** how Doodlebot navigates: it explores (sense → map → move, repeated), then plans the shortest path and rescues Princess-Bot.
3. **Compare** two runs of a fair test that change only the sensor range, and explain the differences using their own measurements.

Why These Concepts

Sensing, mapping and planning are the core of autonomous navigation (AI4K12 Big Ideas 1–2). A fair test confronts the idea *"a better sensor gives a shorter path"*; BFS vs A* is an extension.

Lesson Flow

Time	Facilitator says and does	Students do	Slides
Warm-up (6 min)			
0–1	*"How does a robot vacuum know where your couch is?"* Highlight: it has to sense it.	Turn and talk, share.	1–2
1–5	**Mystery Map:** *"You are Doodlebot's brain. Where should it go?"* Vote, reveal; on turn 3 Princess-Bot appears.	Vote on each move.	3–5
5–6	Welcome page: *"Nobody drives Doodlebot: it is autonomous."*	Listen and look.	6
Concept explanation and gameplay (18 min)			
6–10	Play **How Doodlebot Thinks**. Teach: sensor, map, unknown, shortest path. Part 1: explore (sense, map, move). Part 2: go home, plan, rescue. Check: *"If Doodlebot senses Princess-Bot early, should it rescue her now or finish its map? Why?"* (Finish: a hidden gap might exist.)	Say the two parts; answer the check.	7–9
10–12	*"What if the sensor sees 4 squares instead of 1?"* Don't reveal the answer.	Circle predictions (Part 1).	10
12–21	Experiment (page 2). Check every pair sees Maze ID 1X6Y.	Run sensor 1, then 4; record (Part 2).	11
21–24	Tally results. *"Which number changed? Which stayed the same? Why?"* Reveal the chart and the big idea.	Two students explain.	12–14
Wrap-up and reflection (6 min)			
24–28	Reflection questions; offer sentence starters.	Write (Part 3).	15
28–30	Real robots. A student asks Doodlebot: *"Why did a bigger sensor help you?"*	Listen.	16

The Experiment

1. Build Tools → Ready-made mazes → **Lesson maze**. Check the Maze ID is **1X6Y**.
2. Mission Settings → sensor range **1**, speed **Fast** → **Start Exploring**. Record both metrics.
3. **Try a different sensor range** → **4** → run again and record. Only the sensor range changes.

Metrics and Answer Key

Metric (Lesson maze 1X6Y, BFS)	Sensor 1	Sensor 4
Steps to finish mapping	78	29
Shortest path length (green rescue path)	21	21
Sensed Princess-Bot at step / cells searched	24 / 74	15 / 74

These are reference results for this maze only. If a pair's numbers differ, check the Maze ID, the sensor setting, the algorithm (BFS) and that the run finished.

Reflection Questions

Question	A strong answer
1. Which run finished mapping in fewer steps? Why?	Sensor 4: it senses more squares each step, so it learns the map faster.
2. Was the shortest path different? Why?	No, both 21: the finished map is the same, so the shortest way is the same.
3. What are a real robot's "black squares"?	E.g. a robot vacuum doesn't know where furniture is in a new room.

Evidence of Learning

Use each student's completed worksheet. For each row, choose the level that best matches their work.

What to look at	Beginning	Developing	Secure
Results table (Part 2)	A run is missing, or the runs used different mazes	Both runs recorded, but the Maze ID is not checked	Both runs on Maze ID 1X6Y (78 / 29 and 21 / 21)
Prediction and conclusion (Part 2)	Prediction check left blank	Ticks whether each prediction was right	Also completes the conclusion: steps fell, path stayed the same
Reflection 1 and 2 (Part 3)	Repeats the numbers without a reason	Gives a reason for one of the two results	Explains both results using the words "sensor" and "map"
Reflection 3 (Part 3)	Names a real robot only	Names something that robot cannot know yet	Explains that the robot must sense it before it can plan

Support and Troubleshooting

- **Before class:** npm start, open localhost:3000 in Chrome, print worksheets. Show students how to reload the maze: Build Tools → Ready-made mazes → Lesson maze.
- **Maze edited by accident, wrong Maze ID, "No path" or "Never found her":** reload the Lesson maze, then run again.